ClimaWatch — visual annex

Figure 1. Architecture and research pipeline

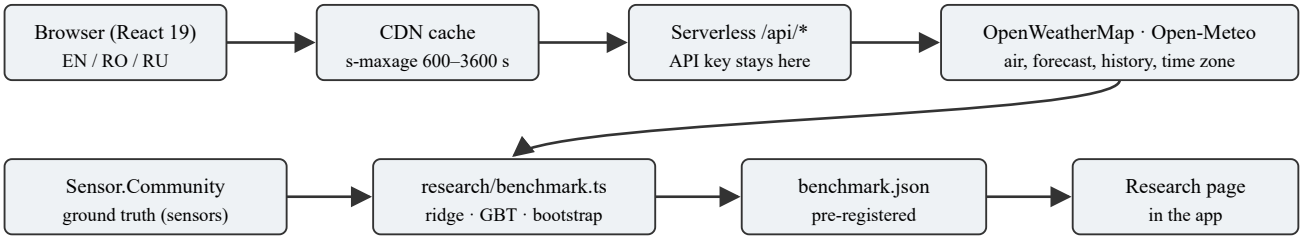


Figure 2. Benchmark: 576 test windows (13,791 hours), blocked cross-validation

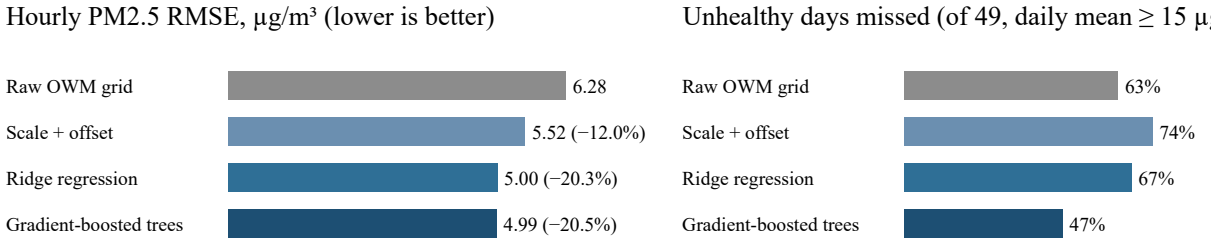


Figure 3. Upstream API calls per month for 10,000 users (4 visits each)



The CDN cache removes 80.1% of the calls; without it the audience exceeds the free quota.

Table 1. Cost and impact summary

Item	Dedicated display monitors	ClimaWatch
Hardware for 100 sites	€31,900 (IQAir AirVisual Pro, list price)	€0 (any browser)
Hosting and API per month	not applicable	€0 inside the free tiers
Users per month inside the free tier	8,333 without cache	119,706 with cache
Avoided asthma emergency episodes per 10,000 users a year	—	0.001 – 0.63 (base 0.017)
Estimated health-system saving per 10,000 users a year	—	€0.22 – €655 (base €6.04)

Health figures are an order-of-magnitude model with a stated assumption for the baseline episode rate (see abstract, section 4). Display prices: IQAir EU list price; PurpleAir Flex US$289 and Touch US$209.